The Normal distribution

STAT 521
Applied Probability and Statistical Inference

Continuous Probability Distributions: Discovery of the Normal Distribution

- **1733 – Abraham de Moivre**: Derived the bell-shaped curve as a large n approximation to the binomial.
- **1770s-1800s – Pierre-Simon Laplace**: Articulated an early Central Limit Theorem; applied the error law to astronomy/physics.
- **1809 – Carl Friedrich Gauss**: Justified the method of least squares via the normal error law (hence 'Gaussian').

Continuous Probability Distributions: Discovery of the Normal Distribution

- **1820s-1830s – Adolphe Quetelet**: Applied the distribution to social measurements (heights, crime rates), popularizing the notion of the 'average man'.
- **Late 19th c. – Francis Galton & Karl Pearson**: Extended use in biology/psychology; formalized statistical methodology with the normal as a cornerstone.
- **Early 20th c.**: Normal distribution became central to inference (sampling distributions, regression, hypothesis testing, ANOVA).

The Normal distribution

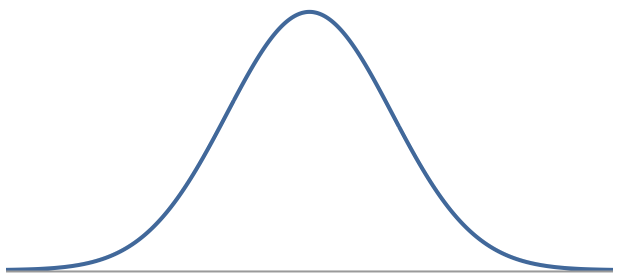
The normal probability density function is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

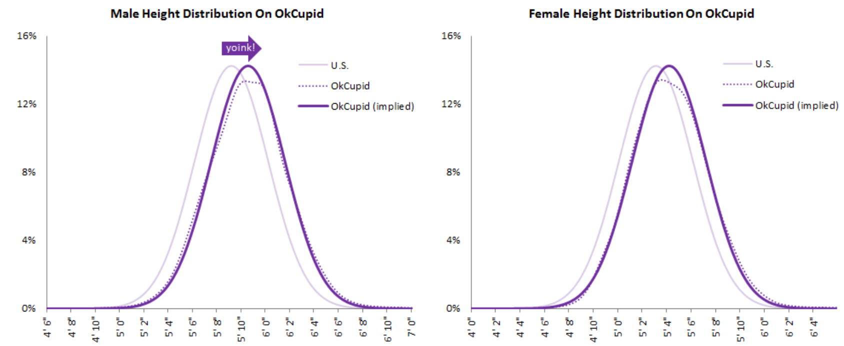
- $N(\mu, \sigma) \to$ Normal with mean μ and standard deviation σ
- It reaches its maximum at $x = \mu$.
- It is symmetric with respect to the line $x = \mu$
- As $x \to +\infty, -\infty$, the curve gets closer and closer to the horizontal axis without ever touching it i.e., the x-axis is the horizontal asymptote).

The Normal distribution

- Unimodal and symmetric, bell shaped curve
- Most variables are nearly normal, but real data is never exactly normal



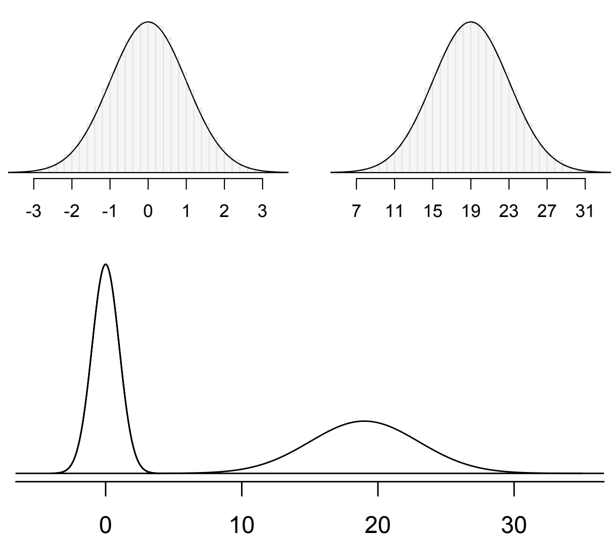
Heights of males and females



http://blog.okcupid.com/index.php/the-biggest-lies-in-online-dating/

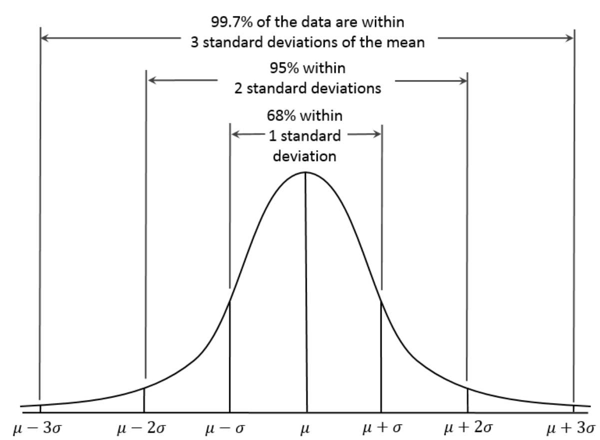
Normal distributions with different parameters

$N(\mu = 0, \sigma = 1)$ $N(\mu = 19, \sigma = 4)$



The Empirical Rule (or the 3-sigma rule)

In a normally distributed data set it is unusual to observe values that are more than two standard deviations away from the mean.



Standard Normal distribution

A very important special case of the normal distributions is the one with $\mu = 0$ and $\sigma = 1$, called this the *standard normal distribution*.

The probability density function is:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad -\infty < z < \infty,$$

and we write $Z \sim \mathcal{N}(0, 1)$.

Any Normal distribution can be standardized by the transformation

$$z = \frac{x - \mu}{\sigma}$$

Conversely, if $Z \sim \mathcal{N}(0, 1)$, then the transformation $x = \mu + \sigma z$ produces $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\sigma > 0$.

Continuous probability distributions in R

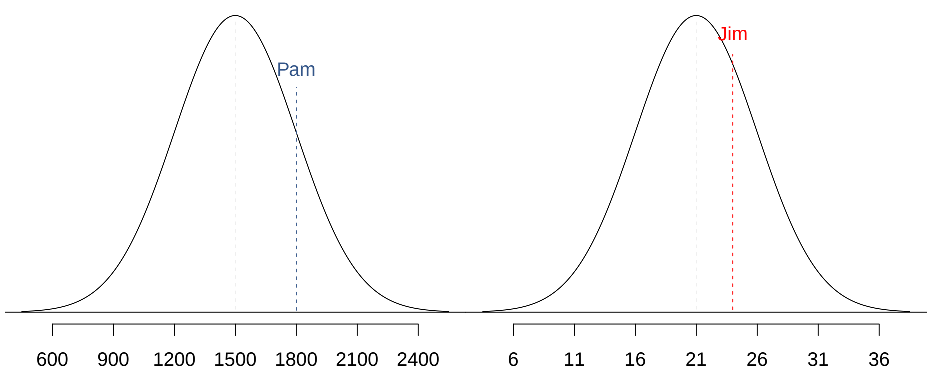
The normal distribution's CDF does not have a closed-form expression in terms of elementary functions.

- *norm* for Normal distribution
- *beta* for Beta distribution
- *gamma* for Gamma distribution

Prefixes *p*, *q*, *d*, and *r* depending on the desired function:

- *p* for "probability", the cumulative distribution function (CDF)
- *q* for "quantile", the inverse CDF for a continuous random variable
- *d* for "density", the probability density function for continuous (PDF)
- *r* for "random", a random variable having the specified distribution

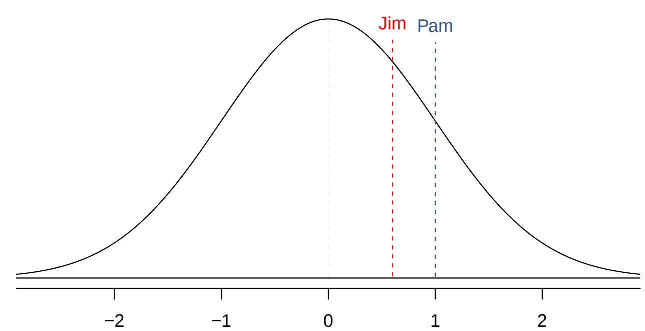
SAT scores are distributed nearly normally with mean 1500 and standard deviation 300. ACT scores are distributed nearly normally with mean 21 and standard deviation 5. A college admissions officer wants to determine which of the two applicants scored better on their standardized test with respect to the other test takers: Pam, who earned an 1800 on her SAT, or Jim, who scored a 24 on his ACT?



Standardizing with Z scores

Since we cannot just compare these two raw scores, we instead compare how many standard deviations beyond the mean each observation is.

- Pam's score is $\frac{1800-1500}{300} = 1$ standard deviation above the mean.
- Jim's score is $\frac{24-21}{5} = 0.6$ standard deviations above the mean.



Standardizing with Z scores (cont.)

- These are called *standardized* scores, or *Z scores*.
- Z score of an observation is the number of standard deviations it falls above or below the mean.

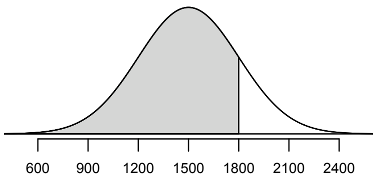
Z scores

$$Z = \frac{observation - mean}{SD}$$

Note: *Z scores can be used to describe observations from distributions of any shape (not just normal) but only when the distribution is normal can we use Z scores to calculate percentiles.*

Percentiles

- *Percentile* is the percentage of observations that fall below a given data point.
- Graphically, percentile is the area below the probability distribution curve to the left of that observation.

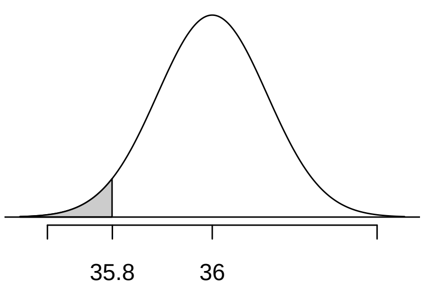


- In R:

```
> pnorm(1800, mean = 1500, sd = 300)
[1] 0.8413447
```

At Heinz ketchup factory the amounts which go into bottles of ketchup are supposed to be normally distributed with mean 36 oz. and standard deviation 0.11 oz. Once every 30 minutes a bottle is selected from the production line, and its contents are noted precisely. If the amount of ketchup in the bottle is below 35.8 oz. or above 36.2 oz., then the bottle fails the quality control inspection. What's the probability that the amount of ketchup in a randomly selected bottle is less than 35.8 ounces?

Let X = amount of ketchup in a bottle: $X \sim N(\mu = 36, \sigma = 0.11)$



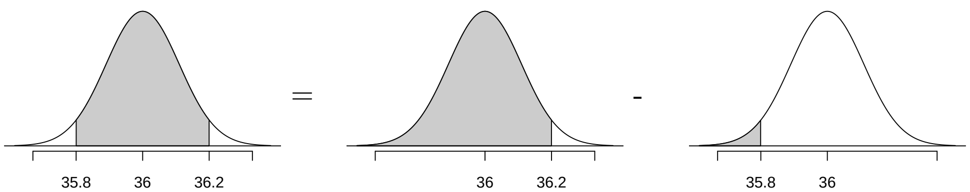
$$Z = \frac{x - \mu}{\sigma} = \frac{35.8 - 36}{0.11} = -1.82$$

$$P(X < 35.8) = P(Z < -1.82) = 0.0344$$

```
> pnorm(35.8, mean = 36, sd = 0.11)
[1] 0.03451817
> pnorm(-1.82) # Standard Normal Distribution
[1] 0.0343795
```

What percent of bottles <u>pass</u> the quality control inspection?

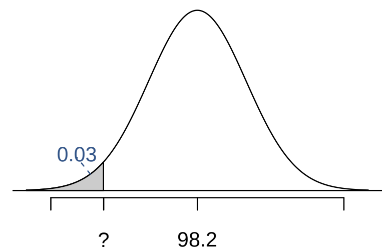
$$P(35.8 < X < 36.2) = ?$$



$$P(X < 36.2) - P(X < 35.8) = 0.9656 - 0.0344 = 0.9312$$

```
> pnorm(36.2, mean = 36, sd = 0.11)
[1] 0.9654818
> pnorm(35.8, mean = 36, sd = 0.11)
[1] 0.03451817
```

Body temperatures of healthy humans are distributed nearly normally with mean 98.2°F and standard deviation 0.73°F. What is the cutoff for the lowest 3% of human body temperatures?



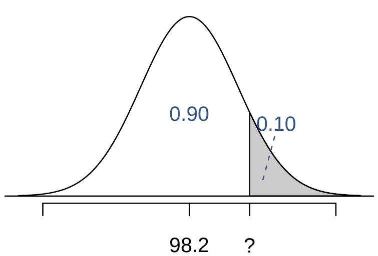
$$P(X < x) = 0.03 \rightarrow P(Z < \textit{-1.88}) = 0.03$$

$$Z = \frac{x - \mu}{\sigma} \rightarrow \frac{x - 98.2}{0.73} = -1.88$$

$$x = (-1.88 \times 0.73) + 98.2 = 96.8$$

```
> qnorm(0.03)
[1] -1.880794
> qnorm(0.03, mean = 98.2, sd = 0.73)
[1] 96.82702
```

What is the cutoff for the highest 10% of human body temperatures?



$$P(X > x) = 0.10 \rightarrow P(Z < \textit{1.28}) = 0.90$$

$$Z = \frac{x - \mu}{\sigma} \rightarrow \frac{x - 98.2}{0.73} = 1.28$$

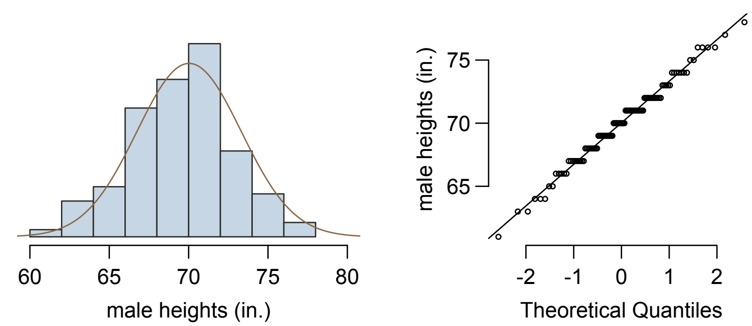
$$x = (1.28 \times 0.73) + 98.2 = 99.1$$

```
> qnorm(0.10, lower.tail = FALSE)
[1] 1.281552
> qnorm(0.10, mean = 98.2, sd = 0.73, lower.tail = FALSE)
[1] 99.13553
```

Checking for Normality

Use a Normal probability plot a.k.a. Quantile-Quantile (Q-Q) plot.

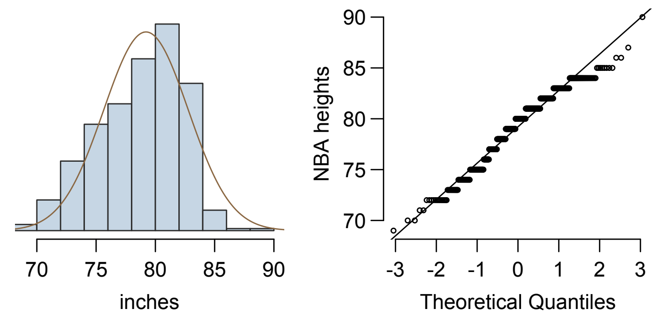
- Example: histogram and *normal probability plot* of a sample of 100 male heights.
- The points appear to jump in increments in the normal probability plot since the observations are rounded to the nearest whole inch.



Anatomy of a normal probability plot

- Empirical quantiles (based on data) are plotted on the y-axis of a normal probability plot.
- Theoretical quantiles (following a normal distribution) on the x-axis.
- If there is a one-to-one relationship between the empirical and the theoretical quantiles, it means that the data follow a nearly normal distribution.
- Since a one-to-one relationship would appear as a straight line on a scatter plot, the closer the points are to a perfect straight line, the more confident we can be that the data follow the normal model.

Below is a histogram and normal probability plot for the NBA heights of $n = 435$ players from the 2008-9 season. Do these data appear to follow a normal distribution?



In base R:

```
> qqnorm(y)
```

Normality in Practice

- In real applied work, exact normality is rarely required.
- Use caution when data are strongly skewed or contain outliers.
- Approximate normality often suffices due to the Central Limit Theorem.
- For large samples, even visibly non-normal data can yield valid inferences.
- For small samples ($n < 30$), visual Q-Q assessment can be misleading.
- Supplement with formal normality tests (Shapiro–Wilk, Anderson–Darling), specially for residuals (not raw data).